



Twin Engine Throttle

Electronics, Firmware & Configuration Guide

Applies to	Details
Throttle models	Felon, Raptor
Firmware	HOTAS Throttle Universal Firmware (02_Throttle.ino)
Configuration app	Tactical MFD Control Hub (Windows application, .exe)
Firmware updates	https://github.com/TheRaelNoisyBoi/DIY-HOTAS-Ecosystem/tree/main
Companion document	The Assembly Instructions of your throttle model (inputs layout, BOM, schematic, mechanical build)

Use this guide after the mechanical assembly is finished. It covers the electronics wiring and I2C addresses, how to verify the wiring, how to flash the firmware and how to configure the throttle with the Tactical MFD Control Hub app (calibration, axis inversion, filters and the ATB mode).

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1.- How the system works

The throttle uses a **hybrid input system**. Buttons and the four main axes go through the I2C bus, while the thumb mini stick connects directly to two analog pins on the Arduino. Everything lives in one unit, controlled by a single Arduino Pro Micro.

Component	I2C address	Function
Arduino Pro Micro (ATmega32U4)	—	Main controller: USB game controller + serial port. Reads the mini stick on A0 / A1
PCF8575 expander #1	0x20	Buttons 1–16
PCF8575 expander #2	0x21	Buttons 17–31
ADS1115 16-bit ADC	0x4A	Rotary Z, right throttle Rx, left throttle Ry, flaps Rz

1.1.- Axis layout

Wire each sensor to the input shown below. The firmware expects exactly this order.

Axis	Control	Sensor	Connects to
X	Mini stick, horizontal	Mini stick (VRx)	Pro Micro A0
Y	Mini stick, vertical	Mini stick (VRy)	Pro Micro A1
Z	Rotary knob	10 kΩ potentiometer	ADS1115 A0
Rx	Right throttle lever	SS49E hall sensor	ADS1115 A1
Ry	Left throttle lever	SS49E hall sensor	ADS1115 A2
Rz	Flaps lever	10 kΩ potentiometer	ADS1115 A3

1.2.- Where your settings are stored

Calibration, axis inversion, filters, jitter threshold and the ATB mode are all stored in the Pro Micro's internal EEPROM. They survive power cycles and firmware updates.

2.- Wiring and I2C addresses

2.1.- Master wiring map

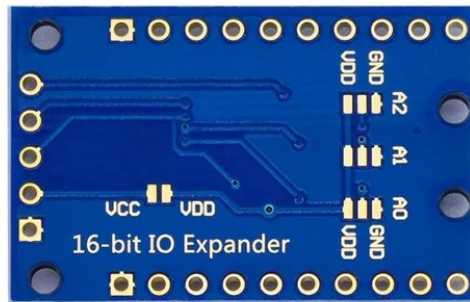
Signal	Pro Micro pin	Connected to
SDA (I2C data)	Pin 2	SDA of both PCF8575 and the ADS1115

Signal	Pro Micro pin	Connected to
SCL (I2C clock)	Pin 3	SCL of both PCF8575 and the ADS1115
Mini stick VRx (horizontal)	A0	Mini stick X output
Mini stick VRy (vertical)	A1	Mini stick Y output
VCC (5 V)	VCC	All modules, hall sensors and potentiometers
GND	GND	All ground points, connected together

The mini stick button (SW) is wired like any other button, to **button 14 (PCF8575 #1, pin P15)**. Appendix B lists the expander pin of every button.

2.2.- PCF8575 button expanders

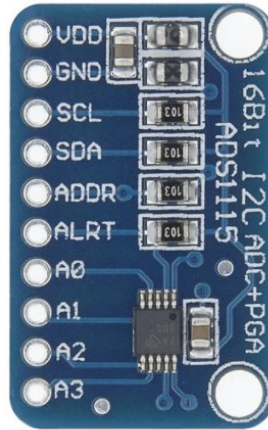
- **Expander #1 – 0x20 (buttons 1–16):** leave it as it came from the factory.
- **Expander #2 – 0x21 (buttons 17–31):** on the back of the board, bridge pad **A0** to **VDD**.



PCF8575 address pads (A0, A1, A2)

2.3.- ADS1115 analog-to-digital converter

The throttle uses address **0x4A**: connect the **ADDR** pin to the **SDA** pin, for example with a short wire on the ADS1115 board itself.



ADS1115: wire ADDR to SDA

Different from the joystick

Do not connect ADDR to GND or VDD. On the throttle it must go to **SDA**, otherwise the firmware won't find the ADC and the four main axes won't move.

3.- Hardware verification (I2C scan)

1. Open `I2C_Scanner.ino` (in the FIRMWARE folder of the official repository, section 4.5) in the Arduino IDE.
2. Select **Tools** → **Board** → **Arduino Leonardo** and the throttle's port.
3. Upload the sketch and open the Serial Monitor at **115200** baud.
4. The output must list: **0x20, 0x21 and 0x4A**.

Result	Likely cause
0x48 instead of 0x4A	ADDR is not connected, or is connected to GND.
0x49 instead of 0x4A	ADDR is connected to VDD instead of SDA.
0x4B instead of 0x4A	ADDR is connected to SCL instead of SDA.
0x20 present, 0x21 missing	The A0 bridge on expander #2 is missing, or both expanders share 0x20.
No addresses at all	SDA/SCL swapped, missing power, or a short on the bus.

4.- Firmware installation and updates

4.1.- Required libraries

Header	Library	How to install
Joystick.h	ArduinoJoystickLibrary (Matthew Heironimus)	Download the ZIP from GitHub → Sketch → Include Library → Add .ZIP Library
PCF8575.h	PCF8575 (Rob Tillaart)	Library Manager
ADS1X15.h	ADS1X15 (Rob Tillaart)	Library Manager
Wire.h, EEPROM.h	Built in	—

Note

Several libraries share the name "PCF8575". Install the one by **Rob Tillaart**; the others use a different API and the sketch will not compile.

4.2.- Custom USB name and VID/PID (do this before the first upload)

By default the throttle shows up in Windows and in games as "Arduino Leonardo". Giving it its own name and USB IDs makes it easy to recognize, and keeps games from mixing up its bindings with other Arduino-based devices. Set them **before** uploading, so you only flash once.

1. Find the AVR core folder that contains `boards.txt`, usually `%LOCALAPPDATA%\Arduino15\packages\arduino\hardware\avr\1.x.x\` (the exact path may vary).
2. In that same folder, create a text file named `boards.local.txt` with the lines below. Values in this file override `boards.txt` without editing it.
3. Restart the Arduino IDE.

```
leonardo.build.vid=0x2342
leonardo.build.pid=0x8039
leonardo.build.usb_product="Twin Engine Throttle"
```

If you also build other Noisy Boi devices

`boards.local.txt` applies to every Leonardo / Pro Micro you upload from this PC. Each device needs its own PID (for example, 0x8038 for the joystick and 0x8039 for the throttle), so edit the file before flashing a different device.

4.3.- Upload the firmware

1. Open `02_Throttle.ino`. Always use the latest version from the FIRMWARE folder of the official repository (section 4.5).
2. Select **Tools** → **Board** → **Arduino Leonardo** and the throttle's port.
3. Click **Upload**.
4. If the upload fails because the port disappears, short **RST** to **GND** on the Pro Micro and click Upload right away (some clones need two quick taps).

After the upload, the throttle shows up in Windows as a game controller (6 axes, 31 buttons, 1 hat) **and** as a serial (COM) port. The app uses that port.

4.4.- Factory defaults

Setting	Default
Inverted axes	None
Alpha 1 (mini stick X, Y)	0.35
Alpha 2 (ADS1115 Z, Rx, Ry, Rz)	0.15
Jitter threshold	15
ATB mode	Off
Calibration	Generic values; calibrate before flying (section 5.5)

4.5.- Firmware updates

Firmware updates, fixes and new features are published in the official GitHub repository:

<https://github.com/TheRaelNoisyBoi/DIY-HOTAS-Ecosystem/tree/main>

1. Open the repository and download the latest throttle firmware from the FIRMWARE folder (**Code** → **Download ZIP**, or open the file and use **Download raw file**).
2. Check the repository notes for changes, new required libraries or new steps before updating.
3. Close the Tactical MFD Control Hub (it uses the same COM port), then upload the new `02_Throttle.ino` exactly as in section 4.3. Check that `boards.local.txt` still has the throttle's PID.
4. Reconnect with the app, press **STATUS** on the LINK page to see the installed version in the log (line Firmware: ...), and check CTRL > INV / FILT / MODE and MON > INPUT.

Your settings are kept

Calibration and settings live in EEPROM, so a normal update keeps them. Recalibrate only if the update notes ask for it, or if the axes behave differently afterwards.

Watch the repository

On GitHub, click **Watch** → **Custom** → **Releases** to get an email when a new version is published.

5.- Tactical MFD Control Hub (configuration app)

The Tactical MFD Control Hub is a Windows desktop app styled as a fighter-jet multifunction display (MFD). With the throttle you can use it to:

- calibrate the six axes,
- invert axes, tune the filters and toggle the ATB mode,
- check every axis, hat and button.

All settings are stored on the throttle itself. **You do not need the app running to fly.**

5.1.- Installation and launch

The app is a single Windows program (Windows 10 or 11, 64-bit). It needs no installer and no additional software.

1. Download Tactical MFD Control Hub.exe from the **Releases** page of the official repository (<https://github.com/TheRaelNoisyBoi/DIY-HOTAS-Ecosystem/releases>) and save it in any folder, for example Documents\HOTAS. You can create a desktop shortcut to it.
2. Double-click it to start. The first start can take a few seconds while Windows unpacks the program.
3. If Windows shows "**Windows protected your PC**", click **More info** → **Run anyway**. The app is not signed with a paid code-signing certificate, so Windows SmartScreen warns about any new program. This happens only the first time.
4. If your antivirus quarantines the file, restore it and add an exception for it. Small independent .exe tools are sometimes flagged as a false positive.

Before connecting

Only one program can use the COM port at a time. **Close the Arduino IDE Serial Monitor** (and any other serial tool) before connecting from the app.

5.2.- Navigating the MFD

Like a real MFD, the screen is surrounded by Option Select Buttons (OSBs). The label next to each OSB shows what it does on the current page. You can click the OSBs, or click directly on the rows, toggles and sliders shown on the screen.

OSB group	Function
Bottom row	Page selection: LINK , CTRL , MON . The active page is boxed.
Left column	Sub-pages (or the port list on LINK).
Right column	Page actions. On CTRL: UP / DN move the cursor, - / + change a slider, SEL toggles or runs the selected row.

The **LINK** flag in the header strip lights up while the app is connected. The GUN, MSL, TD and HFB flags belong to the joystick's haptic features and stay off with the throttle.

5.3.- LINK page – connecting to the throttle



LINK page: port list, connection state and serial log

1. Plug in the throttle and press **SCAN** (bottom-left OSB) to refresh the port list.
2. Select the throttle's COM port with the left OSBs (or click it in the list).
3. Press **CONN**. The app requests the device status.
4. When the throttle answers, **DEVICE** shows **THROTTLE** and the app opens the CTRL page automatically.
 - **STATUS** asks the throttle for its current settings again, including the stored calibration ranges. The reply appears in the serial log.
 - **DISC** closes the connection.
 - Not sure which port it is? Note the list, unplug the throttle, press SCAN, and see which port disappeared.

5.4.- CTRL page – throttle settings

Choose a sub-page with the left OSBs: **INV**, **FILT**, **MODE**, **CAL**. Every change is sent to the throttle and saved immediately; the app then re-reads the status to confirm. The **HFB** sub-page is for the joystick only and shows "HFB AVAILABLE ON JOYSTICK ONLY".

INV – Axis inversion



Toggle **INVERT X / Y / Z / RX / RY / RZ** until every control moves the right way in-game. Check it on MON > INPUT or in Windows' game controller panel. A typical case: if pushing a throttle lever forward lowers the value, invert Rx or Ry.

FILT – Filters

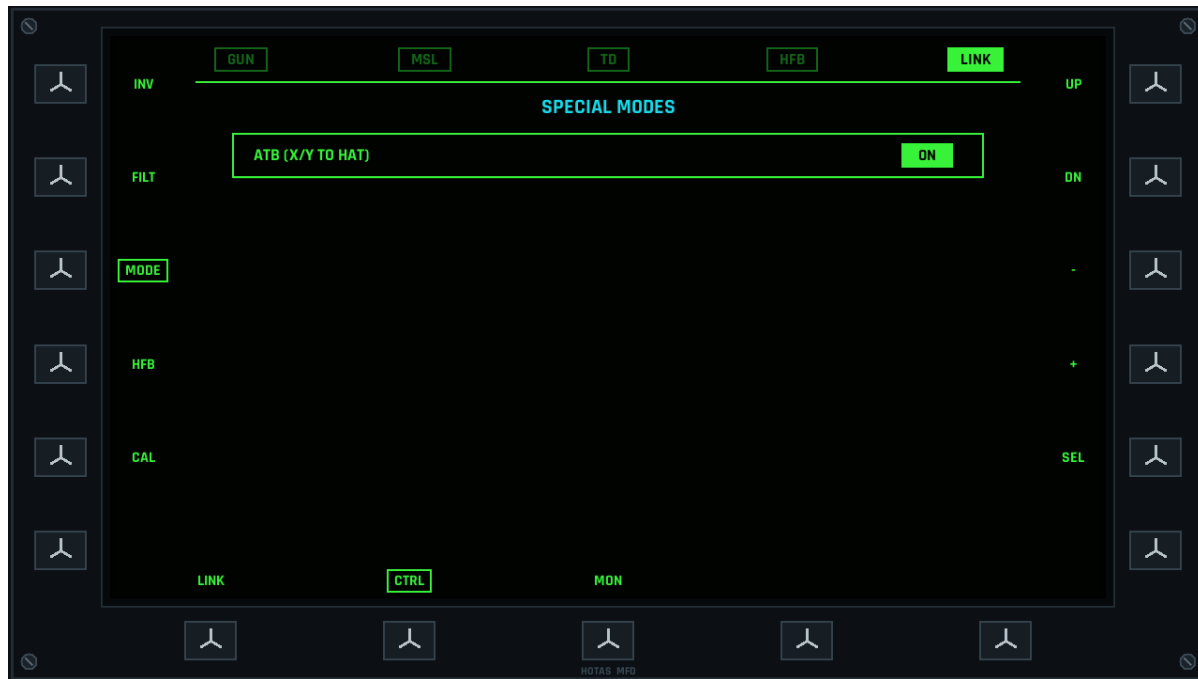


CTRL > FILT

Setting	Range	Default	Effect
ALPHA 1 (ANALOG X,Y)	0.01 – 1.00	0.35	Mini stick filter. Lower = smoother but slower to respond. Higher = more direct, more noise.
ALPHA 2 (ADS Z,RX,RY,RZ)	0.01 – 1.00	0.15	Same, for the throttle levers, rotary and flaps.
JITTER THRESHOLD	0 – 500	15	Minimum change (output range ± 16384) before an axis value updates. Raise it if an axis flickers at rest.

Fast movements are not slowed down: when a lever moves quickly the filter adapts and follows it almost instantly. Drag a slider and release it, or use **- / +**; the value is sent when you release.

MODE – ATB (Axis-To-Button)



CTRL > MODE

ATB (X/Y TO HAT) turns the mini stick into an 8-way hat switch (a direction registers once the stick is pushed past about half of its travel). While it is on, the X and Y axes stay centered so games don't receive both inputs. The setting is stored on the throttle.

5.5.- CAL – Calibration

Calibration teaches the firmware the real minimum and maximum of each sensor, so every axis uses its full range. Calibrate once after the build, and again if you change a sensor, a magnet, a potentiometer or the detents.

1. Connect on the LINK page, then go to **CTRL > CAL**.
2. Select **START CALIBRATION WIZARD**. The throttle starts with the first axis right away.
3. The screen names the axis and shows its live **min / max** values. Move that control slowly to both ends, several times, until the numbers stop changing.
4. Select **SAVE AXIS (NEXT)**. The app refuses to save a range that is too small (less than 20 counts): move the control through its full travel and try again.
5. Repeat for all six axes, in the order shown below.
6. After the last axis the screen shows **CALIBRATION COMPLETE!** and the values are stored.



Calibrating an axis: live min/max values

Step	Axis	Move
1	X	Mini stick fully left and right
2	Y	Mini stick fully up and down
3	Z	Rotary knob from end to end
4	Rx	Right throttle lever from idle to full (past any detent)
5	Ry	Left throttle lever from idle to full (past any detent)
6	Rz	Flaps lever through all positions

Important

All six axes must be calibrated; the throttle has no option to skip or disable an axis. While the wizard runs, the throttle stops sending inputs to the PC. Finish it before flying.

If the app is closed in the middle of a calibration, the throttle keeps waiting. Reopen the app and reconnect (it picks up the wizard where it was), or unplug and replug the USB cable to abort.

5.6.- MON page – monitoring

INPUT – Throttle inputs



MON > INPUT: axes, hat and buttons

Every axis, hat and button of the selected game controller is shown live. Use **JOY- / JOY+** to choose the throttle and **SCAN** to refresh the list. The axes appear as A0–A5 in the order X, Y, Z, Rx, Ry, Rz. The app counts buttons from **0**: button 0 in the app is button 1 in your model's inputs layout. The hat (H0) only moves when the ATB mode is on.

STAT – Status summary



MON > STAT

A one-page summary of the connection and the device settings. Handy when you ask for support: send a screenshot of this page. The DCS page belongs to the joystick's haptic features and is not needed for the throttle.

6.- Troubleshooting

Symptom	What to check
App: "Failed to connect"	Another program has the port open (Arduino Serial Monitor, another app instance). Close it and press CONN again.
Connected, but DEVICE stays "___"	Press STATUS. If nothing appears in the log, re-flash the firmware and check that you selected the throttle's COM port.
Log shows "Error: ADS1115 not connected" at startup	The ADS1115 is not at 0x4A. Check ADDR → SDA and run the I2C scan.
Z, Rx, Ry and Rz don't move	Same as above: the ADS1115 is missing or at the wrong address.
Mini stick X/Y don't move	Check the A0/A1 wires and the mini stick power. Make sure the ATB mode is off.
Two controls are swapped	Compare the wiring with the axis layout (section 1.1).
An axis jitters at rest	Raise the JITTER THRESHOLD, or lower the

Symptom	What to check
	corresponding ALPHA.
Axis moves the wrong way	CTRL > INV.
Axis doesn't reach its ends, or has a dead zone at one end	Recalibrate using the full travel of every control.
Inputs freeze in-game	A calibration is still running on the throttle: finish it, or replug the USB cable.
Games confuse the throttle with another device	Give it its own PID (section 4.2) and rebind it in the game.

Appendix A.- Serial command reference

Everything the app does can also be done from any serial terminal (for example, the Arduino Serial Monitor at **115200** baud with **Newline** line ending). Commands are not case-sensitive.

Command	Action
STATUS	Print the current settings and calibration ranges
CAL	Start the calibration: X, Y, Z, Rx, Ry, Rz in order. For each axis, move it end to end and press Enter (empty line) to save.
INV_X, INV_Y, INV_Z, INV_RX, INV_RY, INV_RZ	Toggle the inversion of that axis
ATB1	Toggle the ATB mode (mini stick → hat)
FIL1 <0.01–1.00>	Set Alpha 1 (mini stick filter)
FIL2 <0.01–1.00>	Set Alpha 2 (ADS1115 axes filter)
JITTER <0–500>	Set the jitter threshold
VCC	Diagnostic: show the measured USB supply voltage (mV)

Appendix B.- Button to expander pin map

Button numbers match the inputs layout in your model's Assembly Instructions. Wire each button between the listed pin and GND.

Button	Expander · pin	Button	Expander · pin
1	#1 (0x20) · P00	17	#2 (0x21) · P00
2	#1 (0x20) · P01	18	#2 (0x21) · P01
3	#1 (0x20) · P02	19	#2 (0x21) · P02
4	#1 (0x20) · P03	20	#2 (0x21) · P03
5	#1 (0x20) · P04	21	#2 (0x21) · P04
6	#1 (0x20) · P05	22	#2 (0x21) · P05
7	#1 (0x20) · P06	23	#2 (0x21) · P06
8	#1 (0x20) · P07	24	#2 (0x21) · P07
9	#1 (0x20) · P10	25	#2 (0x21) · P10
10	#1 (0x20) · P11	26	#2 (0x21) · P11

Button	Expander · pin	Button	Expander · pin
11	#1 (0x20) · P12	27	#2 (0x21) · P12
12	#1 (0x20) · P13	28	#2 (0x21) · P13
13	#1 (0x20) · P14	29	#2 (0x21) · P14
14	#1 (0x20) · P15	30	#2 (0x21) · P15
15	#1 (0x20) · P16	31	#2 (0x21) · P16
16	#1 (0x20) · P17	—	—

Pin P17 on expander #2 is not used.

For questions contact me at: the.noisy.boi@gmail.com

ALL DONE – HAPPY FLYING!